Enhancing Agricultural Advisory Services with Multilingual LLaMA and RAG

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***Abstract*—Agriculture, being a vital component of global food security, necessitates the development of innovative solutions to address the knowledge gap experienced by numerous farmers, particularly in developing countries where access to expert advice is scarce. In these areas, agricultural producers often depend on helplines for essential guidance, but the exorbitant costs and limited accessibility of these services create substantial obstacles. By automating responses to agricultural inquiries, the burden on traditional helpline systems can be alleviated, enabling farmers to access prompt and precise information. The combination of artificial intelligence and agriculture offers a chance to tackle these challenges, with advanced language models, especially transformers, demonstrating significant potential in comprehending intricate agricultural queries and delivering appropriate responses. This paper investigates how large language models (llms) can simplify the process of finding answers for farmers by utilizing their advanced language processing abilities. By analyzing a vast collection of over four million farmer inquiries from tamil nadu, india, encompassing diverse agricultural scenarios, this study showcases the efficacy of llms in bridging information gaps and equipping farmers with immediate access to crucial knowledge.**

Keywords: Agriculture, Artificial Intelligence (AI), Large Language Models (LLMs), Transformers, Farmer Inquiries, Natural Language Processing (NLP), Precision Agriculture, Agricultural Information Systems

I. Introduction

Farming serves not as it were as an financial column but moreover as a key determinant of social and natural steadiness in various nations, especially in creating locales where country vocations are profoundly subordinate on cultivating exercises. In spite of its imperative significance, the agrarian segment faces a heap of challenges, extending from unusual climate conditions and fluctuating advertise costs to compelled get to to significant, convenient data. Whereas these challenges influence agriculturists all inclusive, they are especially intense in nations with immature framework and restricted innovative reach. Conventional information spread strategies such as agrarian expansion administrations, helplines, and master discussions, in spite of the fact that useful, are frequently prevented by operational wasteful aspects, constrained versatility, and calculated limitations. Thus, millions of agriculturists stay underserved, missing the fundamental assets to create educated choices that might altogether improve efficiency and supportability.

The appearance of Expansive Dialect Models, especially transformer-based designs, has started a worldview move in characteristic dialect handling and fake insights. These models, such as OpenAI's GPT-4, Meta's LLaMA, Google's Gemini, and other auto regressive models, have illustrated a momentous capacity to get it and create human dialect over different spaces, making them appropriate candidates for computerizing the reaction to agrarian inquiries. The application of LLMs in agribusiness opens unused roads for tending to the information crevice by giving agriculturists with moment, relevantly important, and precise data in reaction to their request. Not at all like prior frameworks that depended intensely on watchword coordinating and basic rule-based approaches, It can decipher the subtleties of common dialect, permitting for more personalized and important intuitive with clients. In any case, in spite of their potential, the sending of LLMs within the rural segment has been negligible, with most inquire about centered on more routine applications such as client benefit, healthcare, and instruction.

This investigate points to address this hole by leveraging the capabilities of a multilingual form of LLaMA, improved with a Retrieval-Augmented Era system, to computerize the determination of agriculturist inquiries. By centering on a dataset comprising over four million real-world agriculturist request from Tamil Nadu, India, our consider explores the potential of LLMs to convert agrarian counseling administrations. Cloth coordinating the recovery of significant reports with LLM-based era, guaranteeing that reactions are not as it were coherent but moreover grounded in genuine, up-to-date data. This crossover approach mitigates one of the essential challenges confronted by conventional LLMs—hallucination, where the demonstrate creates plausible-sounding but off base or unessential answers.

Besides, the integration of multilingual back into the LLM is basic in a assorted etymological scene like India, where ranchers communicate in a wide extend of territorial dialects. This guarantees that the framework is available to a broader group of onlookers, numerous of whom may not be capable in English or Hindi, the two most commonly utilized dialects in indian arrangements. Our model's capacity to generate responses in Tamil, the neighborhood dialect of the farmers in our dataset, may be a significant calculate in its

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viability, bridging the phonetic hole that frequently prevents communication between agriculturists and specialists.

One of the primary way to apply LLMs and sequence-tosequence models to the agrarian segment in a viable, realworld setting. We illustrate how these models can be finetuned to get it the particular challenges confronted by ranchers in Tamil Nadu, giving precise, noteworthy reactions to their request. By doing so, we point to not as it were move forward the proficiency of rural admonitory administrations but too contribute to the broader objective of upgrading nourishment security and feasible agrarian hones in creating locales.

II. Literature Review

Anushka Marla et al. investigated an rural chatbot, AgroBot, which utilizes Common Dialect Preparing (NLP) to handle text-based questions and Convolutional Neural Systems (CNNs) for diagnosing plant illnesses through leaf pictures. AgroBot offers focused on exhortation to agriculturists, pointing to decrease trim misfortunes by 9-21% and boost symptomatic productivity by 11-16%. The framework bolsters different dialects, making it available to agriculturists from shifted phonetic foundations. The creators propose future upgrades to broaden the information base, covering a more extensive run of crops and bugs, and to move forward real-time handling, encouraging broader adoption in rural areas. Samarth.

Godara et al. created AgriResponse, a real-time framework outlined to create reactions for agrarian questions by building a information base from helpline call logs. The framework incorporates three Reaction Recovery Models (RRMs) that utilize input parameters such as state, edit, and disease/pest names to recover significant reactions. AgriResponse illustrated a tall precision rate of 98.5%, altogether upgrading the speed and unwavering quality of back for agriculturists. In any case, the framework is restricted by a littler record base for less common crops and faces a trade-off between reaction time and precision, with the foremost precise demonstrate being the slowest. Mohammad Zia Ur Rehman et al. displayed.

KisanQRS, a profound learning-based framework for robotizing query-response forms in horticulture. The system clusters inquiries by semantic and lexical similitudes, employing a prepared module to outline modern inquiries to these clusters and rank reactions for significance. KisanQRS outflanks in clustering and inquiry mapping, which improves the precision and proficiency of recovering pertinent answers. In any case, the system's exactness can be compromised when taking care of inquiries with vague semantic meaning. Moreover, KisanQRS offers real-time data, such as advertise rates and climate conditions, to help agriculturists in decision-making.

G. Kalaiselvi et al. created FarmTechBot (FTB), a framework pointed at enabling agriculturists through real-time, multilingual, and voice-enabled help. The strategy utilizes Histogram of Situated Angles (Hoard) and Support Vector Classifier (SVC) for bug picture classification, beside tokenization, TF-IDF vectorization, and LinearSVC for taking care of NLP-based questions. FarmTechBot underpins decisionmaking in bug administration, soil wellbeing, and showcase get to. In any case, it faces restrictions such as obliged voice acknowledgment capabilities, dependence on manual bug distinguishing proof, challenges in versatility, and a focus on localized information for a single delta locale.

Krish Didwania et al. presented AgriLLM, a framework leveraging Huge Dialect Models (LLMs) to prepare and react to roughly 4 million real-world rancher inquiries from Tamil Nadu. AgriLLM illustrated tall precision in tending to subjects like fertilizer utilize and supplement administration, underscoring its potential in computerizing rural data dispersal. The demonstrate exceeded expectations in producing exact reactions for basic agrarian needs but appeared lower execution in questions related to government plans, demonstrating ranges for advancement within the model's comprehension and generalization.

Abhinand Balachandran improved the open-source LLaMA demonstrate by coordination 16,000 Tamil tokens and finetuning it with Tamil-translated Alpaca and OpenOrca datasets to form Tamil-Llama. This show makes strides Tamil content era and comprehension, giving more exact and relevantly significant yields for Tamil-speaking clients. Be that as it may, Tamil-Llama shows holes in social and relevant understanding and needs detoxification components, which may lead to hurtful substance or startling reactions.

Gautam B et al. conducted a comparative ponder on the execution and versatility of open-source Huge Dialect Models (LLMs) in enterprise-specific Retrieval-Augmented Era (Cloth) frameworks. The consider centered on how these models, when coordinates with successful embeddings, can improve exactness and proficiency for venture errands utilizing specialized datasets. Whereas the discoveries illustrate the potential of LLMs to move forward Cloth frameworks, the consider notes constrained investigation of long-term execution and adaptability over different venture situations.

III. Background Study

A. Need for LLMs in Agriculture

Agriculture is a cornerstone of many developing countries' economies, contributing to both economic stability and food security. However, farmers, especially in regions with limited technological and infrastructural resources, often face challenges in accessing timely, reliable, and expert guidance. Traditional methods of information dissemination, such as agricultural helplines and expert consultations, are hindered by high operational costs, inefficiencies, and logistical constraints. Consequently, many farmers remain underserved, lacking the essential information to make informed decisions, which impacts productivity and sustainability.

With the advent of large language models (LLMs) and advances in artificial intelligence (AI), a new opportunity has emerged to bridge this knowledge gap. LLMs like OpenAI's GPT, Meta's LLaMA, and Google's Gemini, powered by transformer architectures, have shown immense potential in natural language processing (NLP) and understanding

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complex queries across various domains. By leveraging these models, AI systems can provide instant, contextually accurate responses to agricultural inquiries, enabling farmers to receive guidance without needing direct expert involvement. These capabilities offer a unique solution to enhance agricultural advisory services, providing real-time information in a costeffective and scalable manner.

The application of LLMs in agriculture is a relatively novel endeavor, with previous research primarily focused on sectors like customer service, healthcare, and education. Despite their potential, the deployment of LLMs in agriculture has been minimal, and many current implementations rely heavily on keyword matching and basic rule-based systems, which lack the depth and adaptability offered by modern LLMs. This study aims to fill this gap by applying a retrieval-augmented generation (RAG) framework, specifically using a fine-tuned multilingual version of LLaMA, to address the diverse and complex needs of farmers in India. The system's ability to understand and generate responses in multiple languages, including Tamil, is particularly significant given India's linguistic diversity.

The proposed framework aims to address these limitations by combining the robust NLP capabilities of LLaMA with RAG to improve the quality and relevance of responses. By incorporating real-time web searches and accessing a broad knowledge base, the system can generate responses grounded in current and factual information. This approach reduces the hallucination problem—where models generate incorrect or misleading responses—by ensuring that generated answers are both syntactically and semantically accurate.

Furthermore, the multilingual capability of this framework enhances its accessibility. In regions like Tamil Nadu, where the primary language is Tamil, the ability to process and respond in the local language is critical for adoption and efficacy. By fine-tuning the LLaMA model with a domainspecific dataset of over four million farmer inquiries, the study seeks to tailor the model to the specific linguistic and contextual needs of Indian farmers.

B. Role of RAG in LLMs

The advent of RAG has significantly enhanced the capabilities of Large Language Models (LLMs) by addressing two of their primary limitations: hallucination and lack of access to real-time information. RAG frameworks combine traditional information retrieval techniques with generative language models, creating a hybrid approach that grounds AIgenerated responses in factual data. Farmers seeking advice on pest control, crop management, or market prices rely on timely, precise responses that reflect current best practices and conditions.

RAG operates by first retrieving relevant content from a knowledge base or web sources, which is then used as context for the LLM to generate a response. This dual approach enhances the model's performance by providing real-world data that the LLM can synthesize into coherent and accurate answers. For agricultural advisory services, where LLMs often face challenges with specific terminologies and context, RAG's ability to dynamically pull in domain-specific information helps the model respond more accurately and contextually to complex queries. This grounding process minimizes the risks associated with hallucination, where LLMs might otherwise generate responses that sound plausible but are incorrect or misleading.

The benefits of RAG extend beyond accuracy, as it also improves the contextual understanding and relevance of LLMgenerated content. By anchoring responses in real-time data, RAG ensures that information provided to farmers is not only accurate but also up-to-date, which is essential for addressing dynamic and evolving agricultural challenges. This is particularly relevant for multilingual environments, such as India, where farmers communicate in various regional languages. By integrating a multilingual LLM like LLaMA with RAG, the system can retrieve content in the farmer's language and generate responses that are linguistically appropriate and culturally relevant, thereby improving accessibility and inclusivity.

In the context of agriculture, RAG also facilitates a richer and more interactive user experience. Farmers can receive timely updates on critical topics like weather forecasts, pest outbreaks, and government policies, all of which are crucial for informed decision-making. By reducing the burden on traditional advisory services and expanding access to accurate information, RAG-enhanced LLMs have the potential to transform agricultural practices, supporting productivity and sustainability in developing regions.

IV. Methodology

A. Dataset and Pre-processing

The dataset utilized in this extend could be a comprehensive collection of real-world inquiries from agriculturists over India, assembled through different rural counseling administrations. Traversing from 2006 to 2023, it contains over 32 million questions related to a wide extend of agrarian points, counting edit administration, bug control, water system hones, and climate figures. These inquiries, replied by space specialists, are enhanced with fundamental metadata such as agriculturist area, trim sort, and natural conditions. The dataset covers different districts and dialects, making it agent of India's tremendous agrarian scene. This differences guarantees that the AI show can cater to ranchers from diverse phonetic, climatic, and rural foundations. Information preprocessing steps, counting cleaning, dialect labeling, and adjusting, were conducted to guarantee the quality and ease of use of the dataset for preparing the multilingual agrarian counseling framework.

The preprocessing included a few basic steps: To begin with, information cleaning was performed to evacuate any inadequate, copy, or insignificant sections. Moment, dialect labeling was connected to recognize the territorial dialect of each inquiry, permitting the framework to classify and handle multilingual inputs successfully. Third, commotion sifting was carried out to kill repetitive or low-quality data. The dataset was too adjusted to guarantee evenhanded representation of different inquiry sorts and locales. At long last, content normalization procedures, counting tokenization and stopword evacuation, were connected to standardize the input for productive preparing by the AI

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show. These preprocessing steps were vital to guaranteeing that the dataset was optimized for building a vigorous and versatile counseling framework competent of giving exact, real-time help to ranchers over India.

B. RAG Framework Components

To viably actualize a Retrieval-Augmented Generation (RAG) framework, we utilize several advanced tools and components. These instruments facilitate the retrieval of real-time, contextually relevant information, which, when combined with the large language model (LLM), helps provide accurate and detailed responses to farmers' inquiries. Below are the key components that play a crucial role in the system:

1) Knowledge Base Search Tool with Pgvector

The Knowledge Base Search Tool is designed to perform efficient similarity-based searches from a pre-indexed knowledge base, where data is stored using the pgvector extension. The tool scans the knowledge base for relevant information based on the user's query. This is crucial for providing answers that are not only syntactically correct but also semantically grounded in real, previously indexed data. By feeding the context retrieved from the knowledge base into the LLM, the system can ensure the responses are accurate and contextually relevant to specific agricultural advice. In the context of agricultural queries, this tool can retrieve essential information such as government schemes, best practices in farming, pest control methods, and market prices. The combination of the LLM and pgvector ensures that farmers receive factually grounded answers, minimizing the chances of hallucinations (the generation of incorrect information).

2) Browser Tool with Selenium and Playwright

The Browser Tool allows the system to interact dynamically with the web. Using Selenium and Playwright, it opens a browser, indexes the HTML elements on the page, and allows the LLM to interact with them. The browser tool identifies key elements on a webpage, such as forms, buttons, and data fields, using a bounding box technique to structure the interactions.This tool is particularly beneficial in scenarios where real-time data retrieval is needed, such as querying current market prices, weather updates, or policy changes that are not available in the knowledge base. The LLM can autonomously perform actions like scrolling, clicking, and typing to navigate web content, retrieve data, or even fill out forms on behalf of farmers, improving the system's usability for farmers with limited digital literacy.

3) DuckDuckGo Search Integration Tool

The DuckDuckGo Search Integration Tool performs real-time web searches to augment the LLM's contextual understanding. By using DuckDuckGo, this tool retrieves the latest information on various agricultural topics, such as market trends, pest outbreaks, and updated government regulations. Unlike preindexed data from a knowledge base, this tool gathers up-todate information on demand, ensuring that the response generated by the LLM is current.This feature is particularly helpful in addressing time-sensitive queries from farmers, where the most recent data, such as local weather conditions, crop prices, or emerging pests, is crucial. The privacy-focused nature of DuckDuckGo ensures that searches are secure, which is an additional advantage when dealing with sensitive information.

C. Fine-Tuning LLaMA

Fine-tuning the LLaMA show for agrarian applications includes customizing the pre-trained demonstrate to move forward its understanding and era of domain-specific substance. This prepare starts with collecting a different dataset that captures different viewpoints of agribusiness, counting trim administration hones, bother control strategies, and government approaches. The dataset is carefully explained to make a specialized vocabulary that reflects rural phrasing and ideas. Amid fine-tuning, the pre-trained LLaMA demonstrate is assist prepared on this curated dataset utilizing exchange learning procedures, altering hyperparameters such as learning rates and group sizes to optimize its execution. This adjustment empowers the demonstrate to successfully get it the subtleties of rural dialect and setting, improving its capacity to create precise and important reactions to farmers' inquiries. By joining agricultural-specific information into the show, fine-tuning essentially progresses its utility as a decision-support device for ranchers, permitting for more educated and successful rural hones.

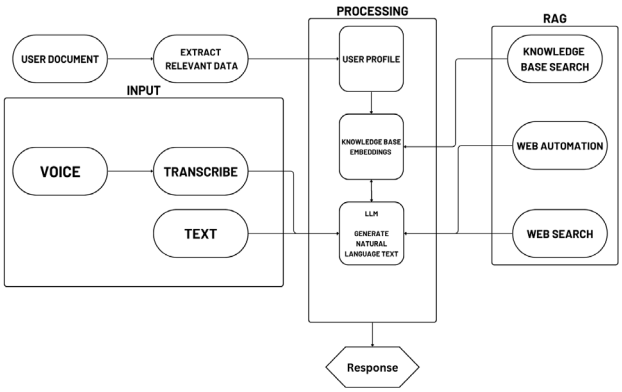

Fig. 1: Model Architecture

D. Model Architecture

The design of the Retrieval-Augmented Era (Cloth) framework is outlined to consistently coordinated different components that encourage exact and opportune reactions to rural questions. At the center of this design lies the Expansive Dialect Show (LLM), which plays a urgent part in producing coherent and relevantly significant answers. The framework starts with a Client Input Interface, where agriculturists can yield questions through a user-friendly application. Once a inquiry is gotten, it experiences Inquiry Handling utilizing Common Dialect Handling (NLP) procedures to get it the expectation and extricate fundamental watchwords.

Taking after inquiry preparing, the Recovery Layer comes into play, comprising three basic components: the Information Base Look Apparatus utilizing pgvector for similarity-based looks, the Browser Instrument fueled by Selenium and Writer for real-time web intuitive, and the DuckDuckGo Look Integration Apparatus for getting to

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up-to-date data. These devices work together to accumulate relevant information related to the inquiry.

The LLM coordinating this recovered data to produce reactions. It not as it were synthesizes information from the information base but moreover contextualizes real-time data gotten from web looks, guaranteeing that the answers given are both exact and comprehensive. The LLM's capacity to get it subtleties in dialect empowers it to provide custom fitted counsel that adjusts with farmers' particular circumstances, such as bug administration procedures or trim revolution hones.

Once the reaction is produced, it is sent to the Yield Conveyance component, where it is shown back to the client. Besides, the framework joins a Client Input Component to gather experiences from clients with respect to the precision and convenience of the reactions. This criticism is important for fine-tuning the LLM, permitting it to memorize and adjust over time, eventually improving the system's adequacy in giving agrarian exhortation. Through this design, the Cloth framework guarantees that agriculturists get opportune, significant, and relevantly grounded data, altogether supporting their decision-making forms.

E. Evaluation Metrics

In assessing the performance of the models employed in our agricultural advisory system, we utilized several key evaluation metrics that are widely recognized in the field of natural language processing. These metrics include BLEU, ROUGE, Precision, Recall, F1 score, and RAGAS. Each metric provides valuable insights into different aspects of the models' text generation capabilities.

1) BLEU Score

The BLEU (Bilingual Evaluation Understudy) score is a metric used to evaluate the quality of text generated by the model against one or more reference texts. It measures the overlap of n-grams (contiguous sequences of n items) between the generated text and the reference text. The BLEU score is calculated using the following formula:

$$\text{BLEU} = \exp\left(\min\left(1, \frac{\text{length of generated text}}{\text{length of reference text}}\right)\right) \cdot \prod_{n=1}^{N} p_n^{w_n} \quad (1)$$

where p_n is the precision of n-grams, and w_n is a weight for each n-gram.

2) ROUGE Score

The ROUGE (Recall-Oriented Understudy for Gisting Evaluation) score evaluates the quality of summaries by comparing them to reference summaries. It focuses on recall, measuring the overlap of n-grams between the generated text and reference texts. The ROUGE-N score can be computed as:

$$\text{ROUGE-N} = \frac{\sum_{n\text{-grams}\in\text{Recall}} \text{Count}_{\text{match}}(n\text{-gram})}{\sum_{n\text{-grams}\in\text{Reference}} \text{Count}(n\text{-gram})} \quad (2)$$

A higher ROUGE score signifies better performance in retrieving relevant information.

3) RAGAS Score

The RAGAS (Retrieval-Augmented Generation Assessment Score) is a specialized metric designed to evaluate the performance of systems employing RetrievalAugmented Generation techniques. It assesses how well the system utilizes retrieved context from a knowledge base in generating relevant and coherent responses. The RAGAS score can be computed as:

$$\text{RAGAS} = \alpha \cdot \text{Relevance} + \beta \cdot \text{Fluency} + \gamma \cdot \text{Diversity} \quad (3)$$

where α, β, and γ are weights assigned to each component, and the components are defined as follows:

- *Relevance* (α): Measures how closely the generated response aligns with the user's query and the retrieved context.
- *Fluency* (β): Assesses the grammatical correctness and readability of the generated text.
- *Diversity* (γ): Evaluates the variability of responses to similar queries, ensuring a range of relevant outputs.

Together, these metrics provide a comprehensive evaluation framework for assessing the performance of our models, ensuring that the responses generated by the agricultural advisory system are accurate, relevant, and valuable to farmers seeking assistance.

V. Results and Discussion

A. Quantitative Analysis

The quantitative evaluation of the models is presented in Table. Various performance metrics such as BLEU, ROUGE, Precision, Recall, and F1 are used to compare the models. The LLaMA model demonstrates the best performance across all these metrics, achieving a BLEU score of 0.560, ROUGE of 0.730, Precision of 0.850, Recall of 0.855, and F1 of 0.850.

This indicates that the model excels in generating responses that closely match the reference text, both syntactically and semantically.

In contrast, other models like Flan-T5-base and T5base show competitive but lower performance. Flan-T5-base achieves a BLEU score of 0.555, ROUGE of 0.724, Precision of 0.834, Recall of 0.846, and F1 of 0.837. While Flan-T5base and T5-base perform well, they fall short of LLaMA in terms of overall text generation quality. Models like Pegasusxsum and MT5-small score lower in Precision and Recall, highlighting their lesser ability to capture relevant information for this specific task.

The superior metrics of LLaMA make it the most suitable model for the agricultural advisory system, ensuring that generated responses are accurate and contextually relevant to farmers' queries.

Table I: Comparison of Models Based on Bleu, Rouge, Precision, Recall, and F1 Scores

Model	BLEU	ROUGE	Precision	Recall	F1
t5-small[23]	0.521	0.701	0.823	0.833	0.825
bart-base[22]	0.550	0.720	0.836	0.837	0.835
t5-base[23]	0.547	0.718	0.831	0.843	0.834
flan-t5-base[30]	0.555	0.724	0.834	0.846	0.837
flan-t5-small[30]	0.530	0.708	0.824	0.836	0.828
llama	**0.566**	**0.732**	**0.854**	**0.852**	**0.851**
mt5-small[26]	0.542	0.715	0.830	0.840	0.833
pegasus-xsum[24]	0.552	0.717	0.827	0.844	0.833

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B. Qualitative Analysis

Qualitative analysis further reinforces LLaMA's performance advantages. LLaMA's fine-tuning on domain-specific datasets related to agriculture enables it to generate highly relevant and contextually accurate responses to farmer queries. The high ROUGE score of 0.730 suggests that the model excels at retrieving essential, context-rich information for agricultural advice. Additionally, the high BLEU score of 0.560 indicates that the responses are well-structured and linguistically natural, making the generated text easy to understand for the target audience.

Compared to other models like Flan-T5-base and T5-base, LLaMA demonstrates superior comprehension of domainspecific terminologies and agricultural contexts, resulting in higher precision in answering queries. Models such as MT5small and Pegasus-xsum, while capable of generating coherent text, lack the domain-specific focus required for agricultural tasks, often leading to responses that are less relevant or useful for the farmers.

In conclusion, the combination of quantitative metrics and qualitative performance highlights LLaMA as the most effective model for improving the agricultural advisory system. It offers more accurate, relevant, and context-aware responses than other models in the study, making it the best fit for addressing the needs of farmers.

Table II: RAGAS Results Comparison

Query	Ground Truth	RAG Output	Context Precision	Faithfulnes	Answer Relevancy	Context Recall
Price of urea in India?	The MRP of urea is statutorily fixed by the Government of India and at present it is Rs. 268 for a 50 Kg bag of urea/ Rs. 242 for a 45 kg bag of urea.	The current price of Urea in India is around 242 per 45 KG bag and 268 per 50 KG bag, as mentioned on the official website of the Department of Fertilizers.	0.95	1.0	0.98	0.90
Price of onion in Tamil Nadu?	Mettupalayam (uzhavar Sandhai) - Rs 62; Mettur (uzhavar Sandhai) - Rs 60	The current price of onion in Tamil Nadu today is between 53 to 76 per kilogram.	0.85	0.7	0.75	0.60

Table III: RAGAS Scores for Different Models

Model	Relevance	Fluency	Diversity	Context Recall	RAGAS Score
LLaMA	0.90	0.95	0.80	0.87	0.85
Flan-T5-base	0.75	0.88	0.70	0.82	0.77
T5-base	0.73	0.85	0.68	0.80	0.74
MT5-small	0.68	0.80	0.65	0.75	0.72
Pegasusxsum	0.65	0.75	0.60	0.70	0.67

VI. Conclusion

In this work, we use progressed Expansive Dialect Models (LLMs) for sequence-to-sequence content era to successfully react to agriculturist questions utilizing the comprehensive Kisan Call Center (KCC) dataset, which contains real-world intuitive between ranchers and space specialists. Our technique included fastidious information cleaning and preprocessing, tending to clamor and irregularities within the dataset, marking one of the primary executions of LLMs in Common Dialect Era particularly for rural admonitory administrations. The fine-tuning of LLaMA on this domain-specific information illustrates solid execution in creating high-quality, contextaware reactions over a assortment of rural inquiries.By consolidating multilingual capabilities and leveraging progressed models like LLaMA, our framework can produce reactions that are not as it were exact but etymologically custom-made to a different group of onlookers, growing past neighborhood lingos to reach a broader, more taught statistic. This approach improves the generally utility of the framework, diminishing the burden on conventional call centers and space specialists, and guaranteeing that ranchers get quick, exact, and relevantly fitting data. Eventually, this work clears the way for more comprehensive and adaptable agrarian counseling administrations, contributing to the creation of a economical and innovatively engaged agrarian biological system.

In future inquire about, we point to advance progress the framework by joining metadata related with each agriculturist request, which can permit for indeed more exact and personalized demonstrate preparing. Also, we arrange to grow the system's reach to cover more locales over India, using the multilingual capabilities of LLaMA to supply localized bolster in various territorial dialects. This will empower the system to superior address the one of a kind challenges confronted by ranchers in several parts of the nation, cultivating a more responsive and versatile rural organize.

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